

Customer Churn Analysis and Prediction

Task 3

Customer Segmentation

Internship Report

Dataset: Telco Customer Churn Dataset

Table of Contents

Table of Contents..... 2

1. Introduction 3

2. Contract Type Revisited 3

3. Segmentation Method 3

4. Churn Rate by Cluster 4

5. Identifying High-Value Customers at Risk 4

6. Business Recommendations..... 5

7. Conclusion..... 5

1. Introduction

This report covers Task 3 of the Customer Churn Analysis and Prediction project: Customer Segmentation. Building on the patterns identified during the exploratory data analysis in Task 2, the objective of this task was to group customers based on tenure, monthly charges, and contract type, analyze churn behavior within each group, and identify high-value customers who are at risk of churning and may need special retention attention.

2. Contract Type Revisited

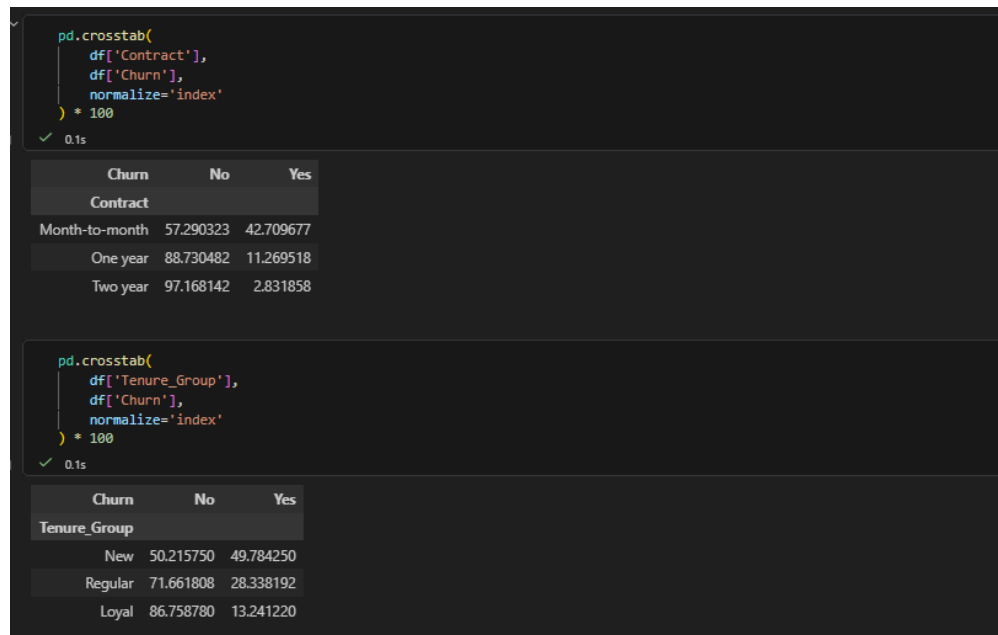


Figure 1. Churn distribution by contract type.

Contract	No Churn	Churn
Month-to-month	57%	42%
One year	88%	11%
Two year	97%	2%

Contract type remains the clearest single risk signal available: month-to-month customers churn at a much higher rate than one-year customers, and two-year customers are essentially stable. This confirms that customers on short-term contracts are significantly more likely to churn, while longer contractual commitments strongly reduce churn risk. From a business standpoint, the company should encourage customers to move to one- or two-year contracts, for example by offering discounts for longer commitments.

Combined with the tenure findings from Task 2, this also confirms that new customers represent the highest risk group: churn is highest in the early lifecycle of a customer, which points to weak onboarding or low initial engagement. Improving the onboarding experience, offering incentives during the first three months, and following up early with new customers are all natural responses to this pattern.

3. Segmentation Method

To move from single-variable analysis to a more holistic customer view, I used K-Means clustering to segment customers into homogeneous groups based on their tenure and monthly charges. I then analyzed the churn behavior within each resulting cluster to identify which segments represent the greatest risk, and which represent the greatest value.

4. Churn Rate by Cluster

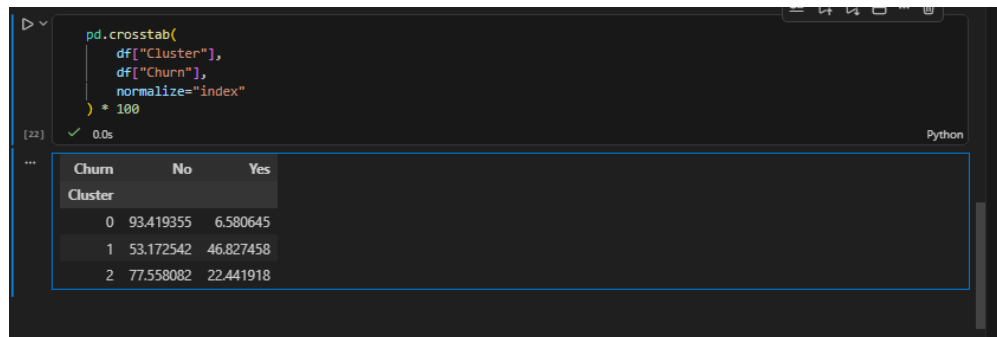


Figure 2. Churn percentage within each customer cluster.

The clustering revealed clear differences between customer segments. Cluster 1 has the highest churn rate (46.83%) — nearly half of the customers in this segment have already left the company — and is primarily composed of month-to-month contract customers, confirming that flexible contracts are associated with higher churn risk. Cluster 0 has the lowest churn rate (6.58%) and is dominated by customers with two-year contracts, indicating strong loyalty. Cluster 2 sits in between, with a moderate churn rate of 22.44%.

Hypothesis confirmed

My working hypothesis was that customers with month-to-month contracts are more likely to churn. The clustering result confirms this directly from the data rather than as a guess: Cluster 1 is mostly month-to-month and has the highest churn, while Cluster 0 is mostly two-year and has the lowest churn.

5. Identifying High-Value Customers at Risk

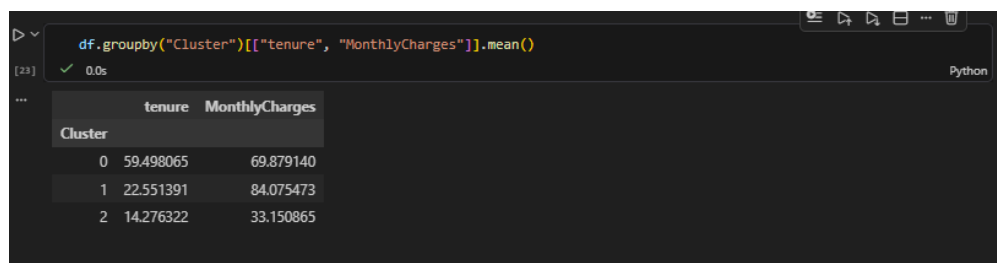


Figure 3. Cluster profiles — average tenure, monthly charges, and churn rate.

Looking at the profile of each cluster in more detail gives a picture of both value and risk:

- Cluster 0 — long-term, low-risk: average tenure of about 60 months, mostly two-year contracts, and a very low churn rate (6.58%), reflecting strong customer loyalty.

- Cluster 1 — high-value, high-risk: the highest average monthly charges (84.08) but also the highest churn rate (46.83%), and mostly month-to-month contracts. These are the most valuable customers in revenue terms, but also the most vulnerable to leaving.
- Cluster 2 — newer, moderate-risk: relatively new customers with lower monthly charges and a moderate churn rate (22.44%).

Based on these profiles, Cluster 1 should be prioritized for retention strategies, because losing these customers would have the greatest financial impact on the company: they pay the most per month, yet they are also the most likely to leave.

6. Business Recommendations

- Prioritize retention offers and proactive outreach for Cluster 1 (high monthly charges, month-to-month contracts).
- Use targeted incentives to migrate Cluster 1 customers toward one- or two-year contracts.
- Continue reinforcing loyalty programs for Cluster 0, since these customers are already low-risk and high-tenure.
- Monitor Cluster 2 customers during their early months, since this is the period where churn risk is still being established.

7. Conclusion

Segmenting customers with K-Means on tenure and monthly charges, and cross-referencing the resulting clusters with contract type and churn, made it possible to move beyond single-variable analysis and identify a specific, high-value, high-risk group of customers (Cluster 1) that should be the primary target for retention efforts. This segmentation directly informs the predictive modeling and business interpretation carried out in Task 4.